

Operating Instructions

1. Make sure that the initial stock values have been entered into registers 'D1000' (raspberry), 'D1010' (lemon), 'D1020' (stabilizer), and 'D1030' (gummy bears).
2. Connect the PLC to the computer and upload the ladder program.
3. Verify that the operating environment is well-lit and that all switches, lamps, and registers are active.
4. Press switch **X007** (Start Workday) – lamp **Y007** will turn on, indicating that the workday has begun.
5. Enter a value into the analog input 'D8280':
 - Value in range **0–5** ← Regular pack.
 - Value in range **6–10** ← Treat pack; lamp **Y001** will turn on.
 - Value in range **5–6 (exclusive)** ← Invalid value; lamp **Y004** will turn on. Correct the value before continuing.
6. Press switch **X001** (Start Prep) – lamp **Y000** will turn on for **5 seconds**, raw materials will be drawn from stock, and the ice cream will be transferred to the prep tanks.
7. Wait for **Y000** to turn off (end of the preparation phase).
8. Press switch **X002** (Cooling) – lamp **Y002** will turn on and cooling will begin. The phase will end when temperature sensor **X003** detects 10°C, or after **8 seconds** (whichever comes first).
9. The mixing phase will start **automatically** – lamps **Y005** and **Y006** will flash alternately (3 seconds for a Regular pack, 5 seconds for a Treat pack). At the end of mixing, the pack counter in 'D1150' will increase by 1.
10. **Important!** Do not press X001 or X002 more than once during the production process of a single pack.
11. To produce another pack – return to step 6.
12. To end the workday – press switch **X005** (Emergency Stop). The analog output 'D8282' will display the number of packs produced and all system variables will be reset.

Troubleshooting

- If lamp **Y004** is on – an invalid value was entered into the analog input (5–6 exclusive). Enter a valid value to clear the error.
- If the workday stops mid-process – one of the raw materials may have run out. Reload stock and press **X007** again.
- For a full system reset – press switch **X005**.